Received: 22 September 2023 | Accepted: 28 February 2024

DOI: 10.1111/jdv.20080

ORIGINAL ARTICLE

JEADV JOURNAL OF THE EUROPEAN ACADEMY OF DERMATOLOGY & VENEREOLOGY

Sun exposure and protection habits: Self-reported attitudes, knowledge and behaviours

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Abstract

Background: As exposure to UV radiation is the primary modifiable environmental risk factor associated with skin cancer, it remains the principal focus of most prevention strategies. Numerous sun protection campaigns have been implemented worldwide; however, their impact on the actual incidence and mortality rates of skin cancer seems to be limited. To create successful skin cancer prevention campaigns, it is important to have a comprehensive understanding of individuals' attitudes and behaviours regarding sun protection. The aim of the current study was to determine and report on the prevalence of self-reported attitudes, knowledge and behaviours regarding two of the major sun protection recommendations—avoidance of sun exposure and use of sunscreens—in an international representative sample across five continents.

Methods: This cross-sectional study was conducted in 20 countries using a web-based online survey.

Findings: A total of 50,552 individuals, comprising 25,388 men (50.22%) and 25,164 women (49.78%), participated in the survey. Among them, 83.2% reported having been voluntarily exposed to the sun (for sun-basking reasons) at least once in the last 12 months, and 47.96% acknowledged being exposed to the sun between the hours of 10 AM and 4 PM. The primary reason for non-adherence was that these hours were the most convenient times (32.28%). Only 24.05% reported applying sunscreen every 2 h when outdoors. Forgetfulness was the primary reason as provided by 27.79% of participants. Males and older age groups were less likely to adopt sun-protective behaviours around the world. Forgetfulness and the challenges posed by time constraints seem to be the biggest barriers to proper adherence.

Interpretation: These findings should prompt the collaboration with health authorities and the manufacturers to enhance adherence by setting reasonable sunscreen prices and creating formulations that make their application less burdensome.

INTRODUCTION

Skin cancer is a growing global concern, with an incidence and prevalence increasing at an alarming rate across the world.[1–3] According to the World Health Organization (WHO), one in every three diagnosed cancers is a skin cancer, and between 2 and 3 million keratinocyte cancers (KC) and 132,000 melanoma skin cancers are diagnosed

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globally each year.[4] One of the key factors contributing to the surge in skin cancer is excessive exposure to ultraviolet (UV) radiation.[5] According to the WHO, the rise in outdoor activities and recreational exposure has played a significant role in the increased incidence of skin cancers in recent decades. Approximately 95% of all skin melanoma cases and deaths in the United States are attributable to UV radiation.[6] Similarly, it has been estimated in several countries that melanomas are among the cancer types with the highest proportion of preventable cases, as UV damage is mostly avoidable.[7,8]

As exposure to UV radiation is the primary modifiable environmental risk factor associated with skin cancer, it remains the principal focus of most prevention strategies implemented worldwide. Different agencies, including the WHO, have provided evidence-based guidelines for sun protection.[9] These include avoiding sun exposure when UV rays are the strongest (between 10 AM and 4 PM); seeking shade; wearing protective clothing, hat and sunglasses; using broad-spectrum sunscreens every 2 h when outdoors; and avoiding sunlamps and indoor tanning. Numerous sun protection campaigns have been implemented worldwide, and have increasingly utilized digital platforms due to their potential wide reach and cost effectiveness.[10–12] While these programs have shown success in reducing the occurrence of sunburns, their impact on the actual incidence and mortality rates of skin cancer seems to be limited.[13,14]

As the incidence of skin cancer continues to rise globally, it is crucial to intensify efforts aimed at promoting and encouraging individuals to adopt protective behaviours that reduce their risk of skin cancer. To create successful skin cancer prevention campaigns, it is important to have a comprehensive understanding of individuals' attitudes and behaviours regarding sun protection. By elucidating the reasons behind non-compliance, we can develop tailored interventions that address the specific barriers individuals' face, with the ultimate aim of promoting widespread adoption of sun protective measures use and reducing the burden of UV radiation-related diseases.

The aim of the current study was to determine and report on the prevalence of self-reported attitudes, knowledge and behaviours regarding two of the major sun protection recommendations—avoidance of sun exposure and use of sunscreens—in an international representative sample across five continents.

METHODS

Study population

This multinational, cross-sectional study was conducted in 20 countries using a web-based online survey by a polling company (HC Conseil Paris, France) between January and April 2023. Institutional review board approvals were not required since the study did not involve any clinical examination and used anonymized data.

For each of the 20 countries (Table 1), the study recruited a representative sample of the general population over 16 years of age. The selection of participants aimed to ensure national representativeness by considering factors such as age, gender, administrative region, environment (including large cities, towns and rural areas) and income distribution in each country, as previously described in previous studies.[15,16] The survey covered the population of 20 countries representing more than 50% of the world's inhabitants. Specifically focusing on individuals aged 16 years or above, the total population from which the sample was derived amounted to 3,213,401,122.

Information regarding the sociodemographic composition of the population in each country was obtained from the Eurostat database, which was updated to reflect the most recent data available during the survey in 2020. Additionally, supplementary data from the United Nations, Department of Economic and Social Affairs, Population Division were also incorporated. World Population Prospects: the 2019 Revision. https://population.un.org/wpp/.

Data were collected via the Internet from a randomly chosen subset of 50,552 individuals from the pool of 900,000 Internet users aged 16 and above and willing to take part in the surveys. Each participant was initially contacted via email, and if contact could not be established, another individual with similar characteristics was randomly selected as a potential replacement.

Questionnaire

The participants were requested to fill out a comprehensive digital questionnaire that consisted of a structured set of questions covering a range of sociodemographic aspects, such as gender, age and other relevant factors. In countries where it was legally permissible, an inquiry regarding ethnic origin was included. Participants were asked about voluntary sun exposure for sun-basking reasons in the past 12 months, sun exposure between 10 AM and 4 PM, and the use of broad-spectrum sunscreens. Subsequently, each participant was queried about the reasons behind their adherence or non-adherence to both sets of recommendations.

Statistical analysis

The 20 countries were grouped into five regions: North America and Australia (Australia, Canada and United States), Latin America (Brazil and Mexico), Europe (Denmark, France, Germany, Italy, Israël, Poland, Portugal and Spain), Africa (Kenya, Senegal, South Africa and United Arab Emirates) and Asia (China, India and South Korea). These five regions were defined and consolidated during a meeting involving three of the authors (HL, CBa and CT.).

In each country, the total population of individuals, as well as the total population by sex and age, were calculated

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TABLE 1 Total patient population.

	Country	Total number of inhabitants (in 2021)	Number of inhabitants 16 years and older (in 2021)	Number of men 16 years and older (in 2021)	Number of women 16 years and older (in 2021)	Proposed sample sizes per country
North America and Australia	Australia	26,439,103	21,372,242	10,528,875	10,843,367	2000
	Canada	38,781,280	32,389,906	16,005,710	16,384,196	2500
	USA	334,319,658	267,302,837	131,408,588	135,894,248	5000
Latin America	Brazil	216,422,436	170,025,851	82,582,233	87,443,618	4001
	Mexico	128,455,559	95,440,511	45,893,170	49,547,341	2500
Asia	China	1,425,671,344	1,169,958,264	590,459,533	579,498,731	5000
	India	1,428,627,653	1,047,596,423	538,029,159	509,567,265	3000
	South Korea	51,784,047	45,509,238	22,616,052	22,893,186	2500
Europe and Israël	Denmark	5,910,901	4,899,008	2,421,671	2,477,338	1000
	France	64,756,570	52,936,023	16,005,710	16,384,196	4000
	Germany	83,294,620	70,881,658	34,733,106	36,148,552	4000
	Israel	9,174,512	6,467,227	3,190,057	3,277,170	1000
	Italy	58,870,750	51,118,409	24,738,447	26,379,963	4001
	Poland	41,026,055	34,458,521	16,512,116	17,946,405	2500
	Portugal	10,247,592	8,809,552	4,101,985	4,707,567	1000
	Spain	47,519,618	40,540,987	19,699,164	20,841,823	4000
Africa	Kenya	55,100,574	33,325,569	16,374,936	16,950,633	500
	Senegal	17,763,154	10,071,911	4,942,551	5,359,981	300
	South Africa	60,414,485	42,300,029	20,234,950	22,065,079	1000
	UAE	9,516,860	7,996,954	5,799,524	2,197,430	750

by direct extrapolation of the proportionally stratified sample proportions in each country.

Descriptive characteristics of the study population were conducted, and qualitative variables were reported as the number (percentage) whereas quantitative variables were reported as the median and/or the mean SD. Participants were grouped into three age categories: 16–34 years, 35–54 years and ≥55 years. Sun protection behaviours and demographic characteristics were compared using the chi-squared test or the Fisher's exact test as needed.

All analyses were conducted in RStudio using R 4.1.3.

RESULTS

Among the participants included in the survey, there were a total of 50,552 individuals, comprising 25,388 males (50.22%) and 25,164 females (49.78%).

Prevalence of sun exposure

Among study participants, 83.2% ±0.33% [81.97%, 82.63%] reported having been voluntarily exposed to the sun (for sun-basking reasons) at least once in the last 12 months, with higher rates in males (85.04% ±0.44% [84.60%, 85.48%]) compared with females (82.38% ±0.47% [81.91%, 82.85%]) ($p < 0.001$). This prevalence was highest in Africa (90.97% ±1.11% [89.86%, 92.08%]), followed by Latin America (88.07% ±0.78% [87.29%, 88.85%]), Asia (85.03% ±0.68% [84.35%, 85.71%]), North America/Australia (78.02% ±0.83% [77.19%, 78.85%]) and Europe (71.62% ±0.60% [71.02%, 72.22%]).

The prevalence of sun exposure was highest in the 16- to 34-year group (87.51%), with a gradual decrease in prevalence as age increased (82.41% and 77.51% in the 35- to 54-year and ≥55 years age groups, respectively) ($p < 0.001$). The higher prevalence of sun exposure among males compared with females and the trend of higher sun exposure among the younger age groups were consistently observed across all five regions (Figure 1).

A total of 47.96% of the participants acknowledged being exposed to the sun between the hours of 10 AM and 4 PM, with higher rates in males (49.33%) compared with females (46.53%) ($p < 0.001$). This prevalence was highest in Africa (60.68%), followed by North America/Australia (57.10%), Asia (46.71%), Latin America (46.55%) and Europe (45.53%). The highest prevalence of sun exposure during these hours was observed in the 16- to 34-year age group (53.75%), with a gradual decrease in prevalence as age increased (45.08% and 41.18% in the 35- to 54-year and ≥55 years age groups, respectively) ($p < 0.001$). Once again, the higher prevalence

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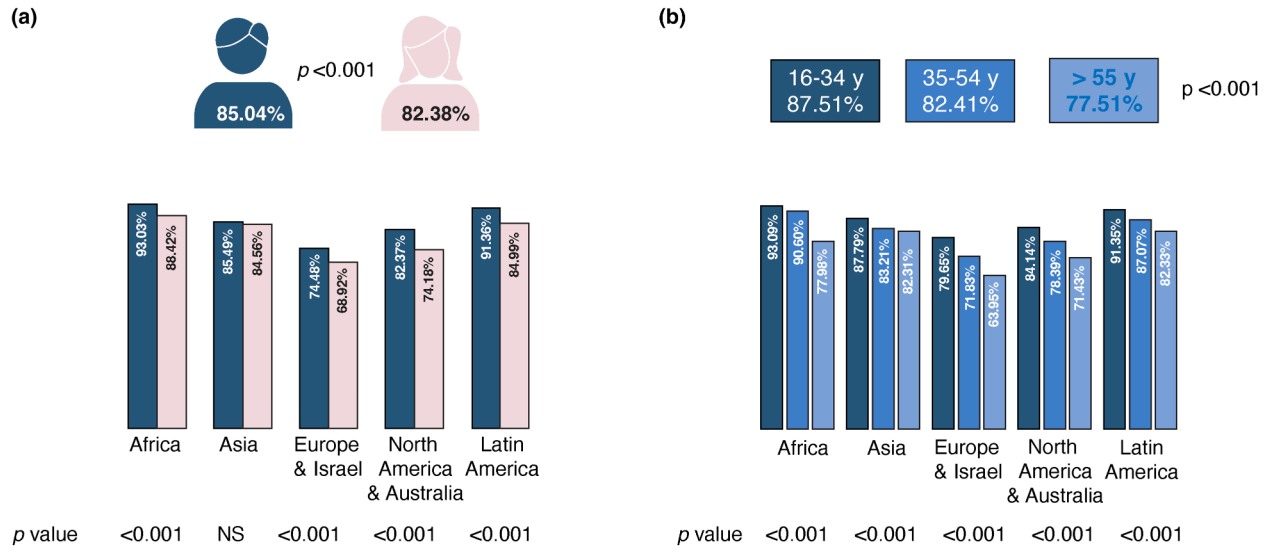


FIGURE 1 Prevalence of sun exposure in the last 12 months according to (a) sex and (b) age.

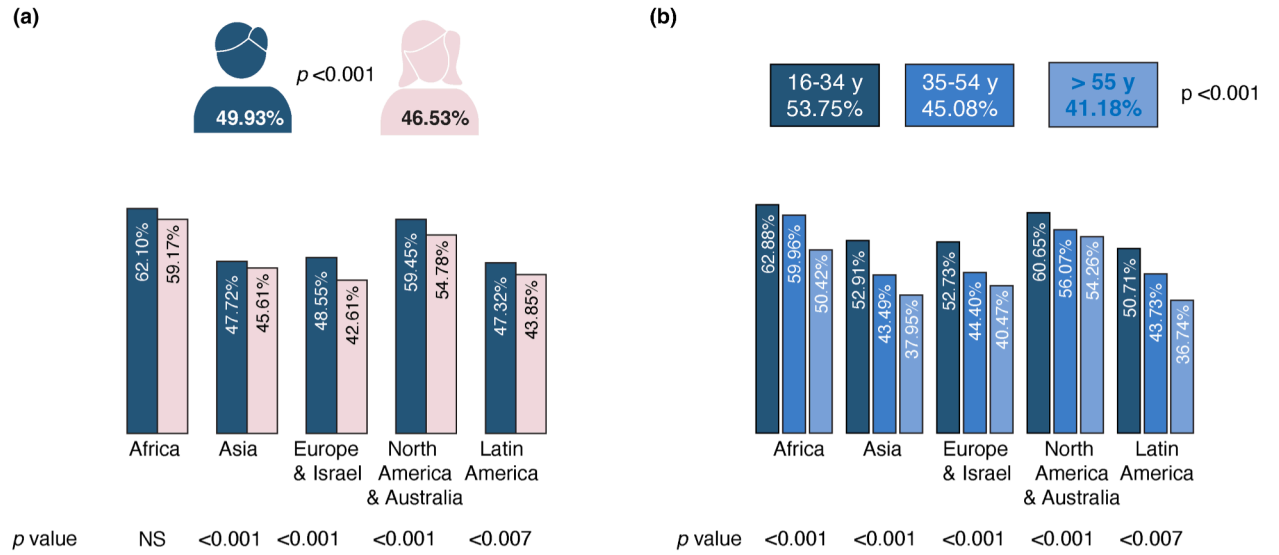


FIGURE 2 Prevalence of sun exposure in the last 12 months between 10 AM and 4 PM according to (a) sex and (b) age.

of sun exposure between 10 AM and 4 PM among males and the younger age groups was consistently observed across all five regions (Figure 2).

Underlying reasons contributing to the failure to avoid sun exposure between 10 AM and 4 PM

The primary reason was that these hours corresponded to the most convenient times (32.28%), followed by the fact that these are the hours when they are available (25.68%) and these are the most pleasant hours (17.88%). Additional reasons included a lack of belief in the prevention messages (5.24%), a lack of comprehension of the prevention message (4.62%) and an uncertainty regarding their own motivations for engaging in such behaviour (2.21%). The distribution of responses to this question based on sex is detailed in Figure 3.

Prevalence of sunscreen application every 2 h when outdoors

Out of the study participants, only 24.05% reported applying sunscreen every 2 h when outdoors. This prevalence was highest in the Latin America regions (29.98%), followed by Asia (25.08%), Europe (22.13%), North America/Australia (16.48%) and Africa (13.93%). Females demonstrated a significantly higher prevalence (28.28%) of this behaviour compared to males (19.87%) ($p < 0.001$). The higher prevalence in female participants was observed across all five regions (Figure 4). The prevalence of applying sunscreen every 2 h was highest in

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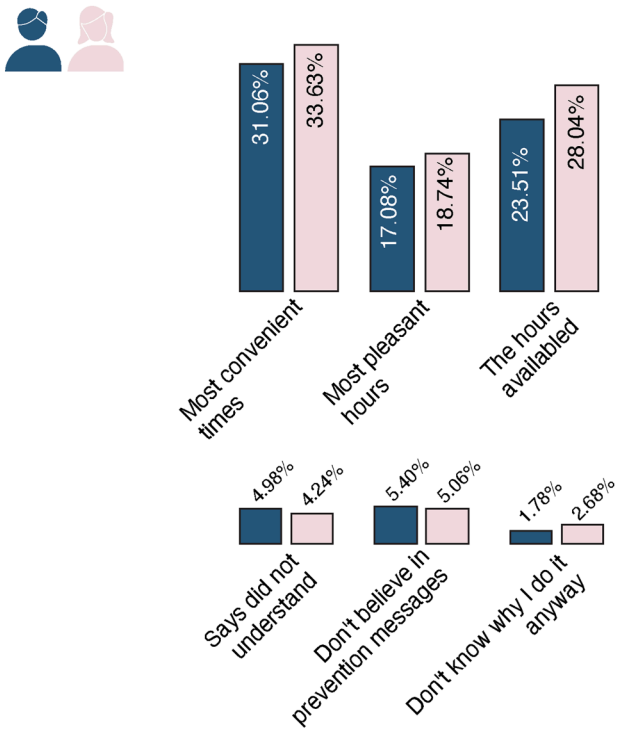


FIGURE 3 Underlying reasons contributing to the failure to avoid sun exposure between 10 AM and 4 PM according to sex.

the 16- to 34-year age group (27.04%) and exhibited a gradual decline with increasing age (25.88% and 12.67% in the 35- to 54-year and ≥55 years age groups, respectively) ($p < 0.001$).

Underlying reasons contributing to the failure of applying sunscreen every 2 h when outdoors

Participants who indicated not applying sunscreen every 2 h when outdoors (75.95%) were queried about the reasons behind their behaviour. As reported by 27.79% of participants, the primary reason provided was that they simply did not think of applying sunscreen every 2 h ('it does not occur to me').

Additional reasons mentioned included finding the task too tedious (18.59%), perceiving sunscreen products as too costly (17.63%), considering them ineffective (16.64%) and lacking awareness of this specific recommendation (15.13%). The distribution of responses to this question based on regions and sex is detailed in the Figure 5.

There was a higher percentage of males compared to females who reported: not thinking of applying the sunscreen every 2 h (28.41% vs. 26.87%), not finding it useful (18.27% vs. 14.20%) and not being aware of this recommendation (16.43% vs. 10.48%). A higher percentage of females (18.93%) perceived sunscreen products as too costly as compared to males (16.54%). Africa exhibited the highest percentage of participants reporting that sunscreen was too expensive (34.40%), and not being aware of this recommendation (23.39%).

Underlying reasons contributing to the practice of sun protection

The prevailing incentive was to avoid sunburn (53.38%), followed by the desire to protect against accelerated skin ageing (40.21%), skin tumours prevention (27.56%), spending more time in the sun (21.19%) and considerations based on a history of sunburn (17.51%). The distribution of responses to this question based on age, sex and region is detailed in the Figure 6.

Protection against accelerated skin ageing was reported by a higher percentage of females (44.13%) compared to males (35.77%) and was mostly a concern of the 35- to 54-year age group (41.52% vs. 40.27% and 37.07% in 16- to 34-year and ≥55 years age groups, respectively, $p < 0.001$%).

DISCUSSION

To our knowledge, this is the first and largest population-based study examining the prevalence of self-reported attitudes and behaviours regarding two of the major sun protection recommendations in an international representative sample across five continents. The study we are reporting on herein covers a representative sample of 20 countries, representing 50% of the world's inhabitants and leading to a total of 50,552 participants aged ≥16 years.

Almost half of the participants (48%) acknowledged being voluntarily exposed to the sun between the hours of 10 AM and 4 PM. Among these individuals, almost a third reported that these were the most convenient times (32%), and a quarter reported that these are the hours when they are the most available (26%). While the timing of lunch breaks can vary depending on cultural norms, workplace policies and individual schedules, in many countries, they typically occur around midday, between 12:00 PM and 1:00 PM. One potential solution could involve structural strategies such as implementing scheduling processes that assign outdoor events to periods of the day when UV radiation is minimal or providing shade or indoor alternatives for such events.

Strong evidence suggests that regular sunscreen use decreases the risks of melanoma and keratinocyte carcinoma.[17,18] The overall consensus is to apply a broad-spectrum sunscreen 15–20 min before exposure, and reapply every 2 h when outdoors.[19] Yet in our study, more than three quarters (76%) of participants did not adhere to the recommended practice of applying sunscreen every 2 h. More than a quarter of these individuals cited forgetfulness as the primary reason for not applying sunscreen. This highlights the need to enhance sun protection campaigns to improve people's overall recall. A randomized controlled trial (RCT) was conducted back in 2007 on 70 volunteers from the general community in Boston USA to evaluate whether using text-message reminders would improve sunscreen use. At the end of the 6 weeks study period, the 35 participants who did not receive reminders had a mean daily adherence rate

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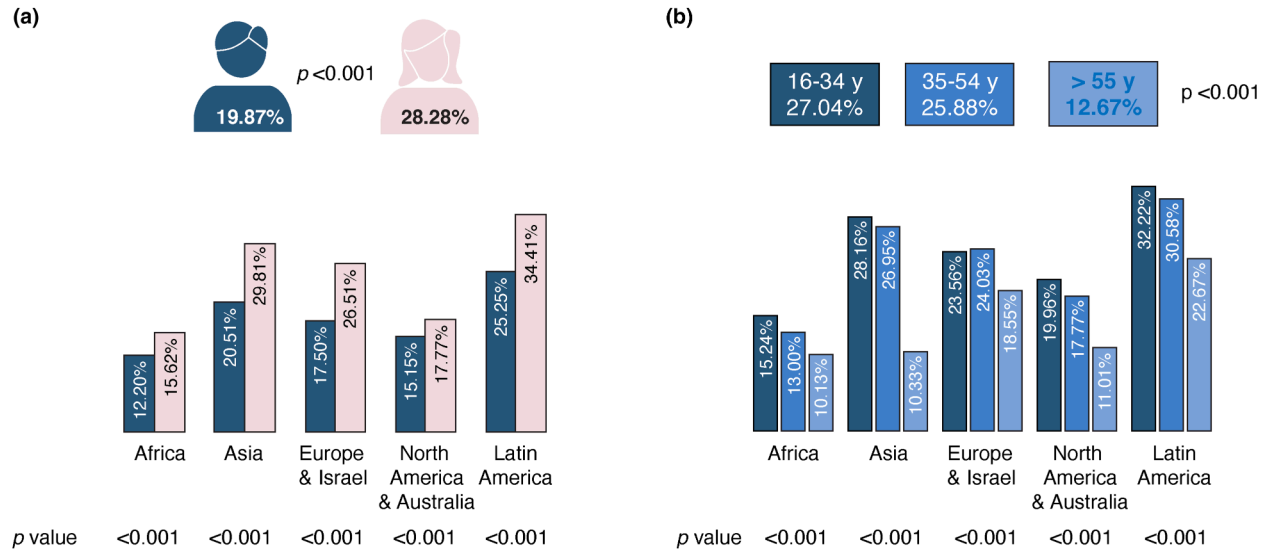
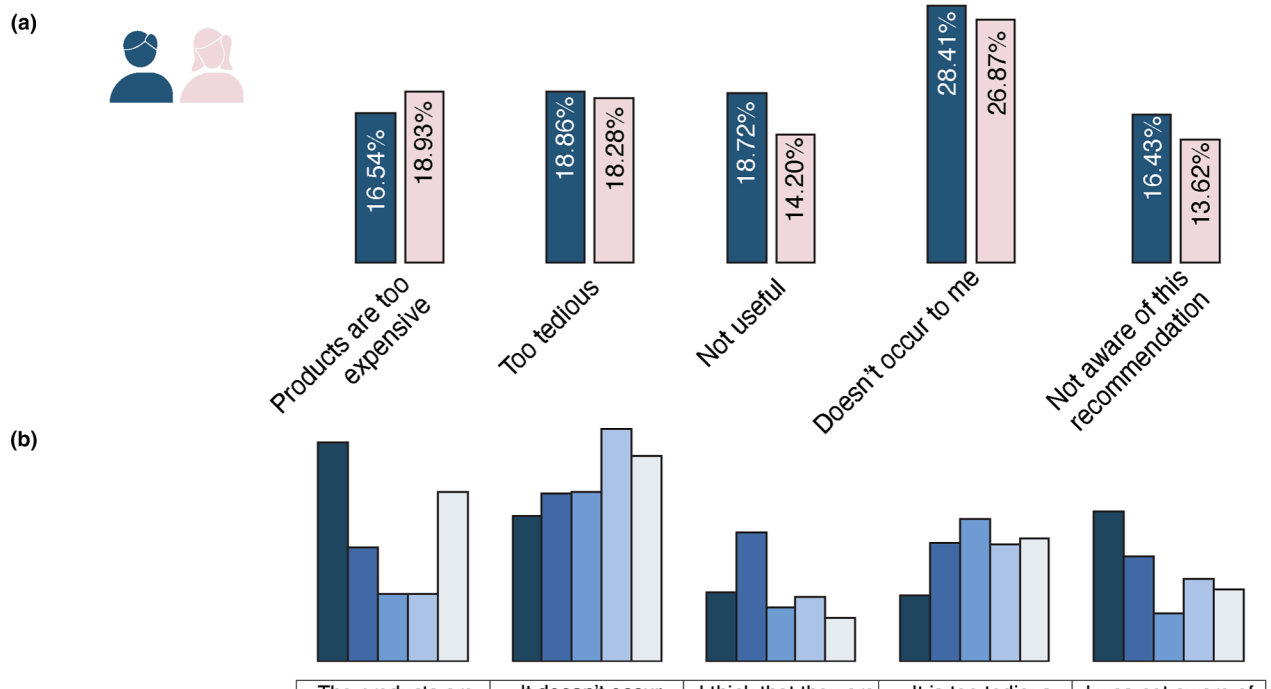


FIGURE 4 Prevalence of sunscreen application every 2 h according to (a) sex and (b) age.

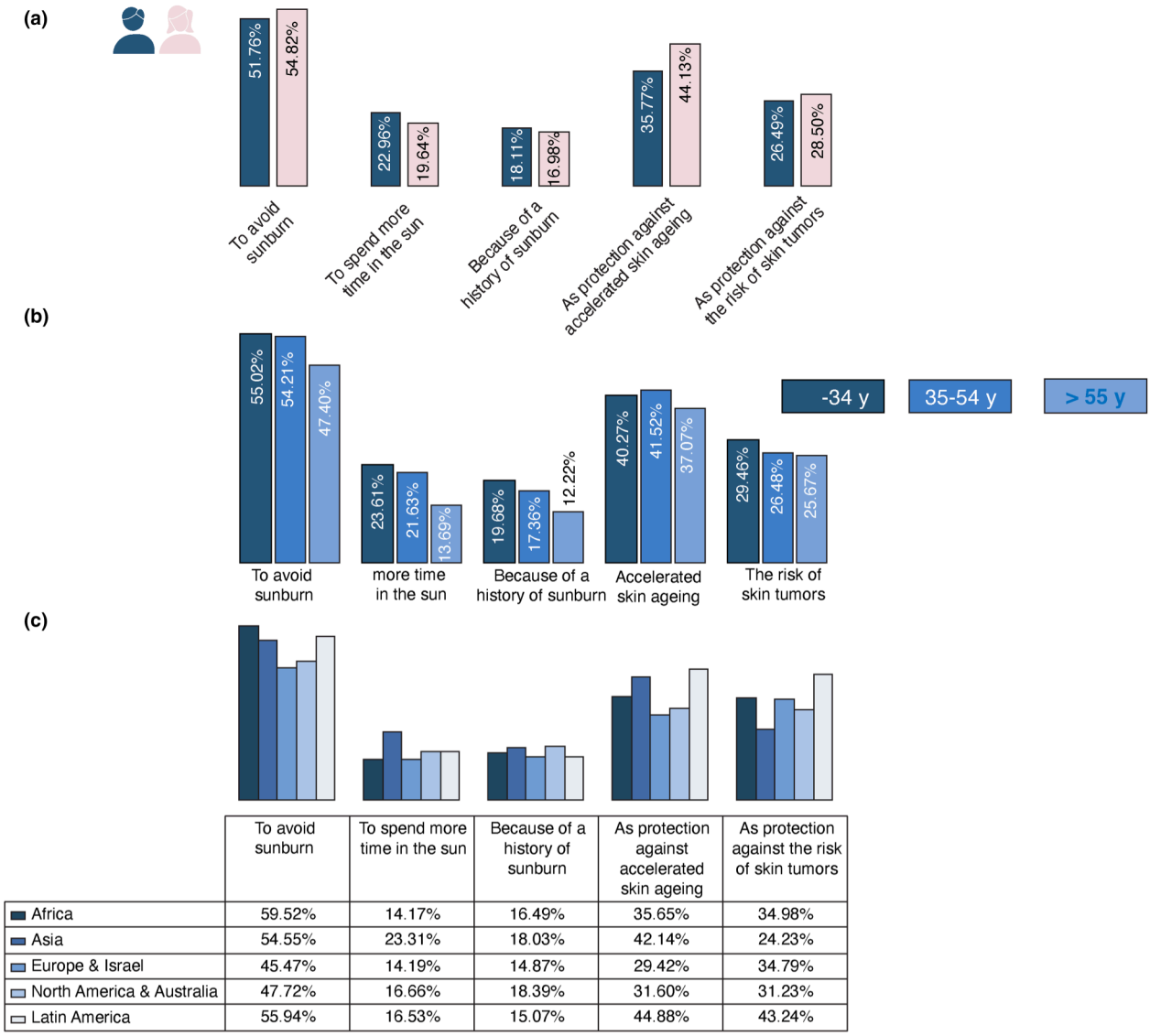


	The products are too expensive	It doesn't occur to me	I think that they are not useful	It is too tedious	I was not aware of this recommendation
Africa	34.40%	22.87%	10.67%	10.37%	23.39%
Asia	17.90%	26.18%	20.08%	18.50%	16.48%
Europe & Israel	10.47%	26.56%	8.32%	22.19%	7.39%
North America & Australia	10.45%	36.50%	10.00%	18.31%	12.90%
Latin America	26.42%	32.18%	6.79%	19.29%	11.25%

FIGURE 5 Underlying reasons contributing to the failure of applying sunscreen every 2 h when outdoors according to (a) sex and (b) region.

of 30% whereas the 35 participants who received daily text messages reminders had a mean daily adherence rate of 56% ($p < 0.001$).[20] A systematic review on 5 RCTs that assessed effects of SMS text reminders; however, did not find a significant benefit on sunburn, sunscreen use and sun protection habits.[21] Nevertheless, utilizing platforms like social media nowadays could offer a potentially more effective approach to reach a wider audience and increase awareness about the importance of sunscreen application.[22] Indeed, social media platforms provide more sophisticated tools for targeted

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	To avoid sunburn	To spend more time in the sun	Because of a history of sunburn	As protection against accelerated skin ageing	As protection against the risk of skin tumors
Africa	59.52%	14.17%	16.49%	35.65%	34.98%
Asia	54.55%	23.31%	18.03%	42.14%	24.23%
Europe & Israel	45.47%	14.19%	14.87%	29.42%	34.79%
North America & Australia	47.72%	16.66%	18.39%	31.60%	31.23%
Latin America	55.94%	16.53%	15.07%	44.88%	43.24%

FIGURE 6 Underlying reasons contributing to sun protection according to (a) sex, (b) age and (c) region.

advertising and customization, allowing campaigns to tailor messages to specific demographics, locations or interests, with the possibility of sponsoring content during specific hours (between 10 AM and 4 PM for example).

Another important aspect to consider is the impact of pricing, as it has a significant influence on sunscreen application. Roughly one in five individuals reported not applying sunscreen due to cost, particularly in Africa where cost was three times more likely to be reported as a reason for not applying sunscreen compared to Europe and North America. The overall economic conditions and average income levels in a particular country or region can influence the affordability of sunscreen. While Africa is a diverse continent with significant variations in economic development, Africa has a higher prevalence of poverty compared to other regions. Poverty affects individuals' purchasing power and their ability to afford various goods and services, including sunscreen. While studies performed in Europe and North America conclude that sunscreens are generally affordable,[23,24] it is worth noting that there are national initiatives to provide more affordable or even free sunscreens.[25,26] Efforts should also be geared towards increasing access to affordable sunscreen in regions with higher sun exposure, including Africa and Latin America.

Our study results are consistent with prior reports indicating that males are less likely to adopt sun-protective behaviours.[27,28] Sun protection messaging frequently often targets females and emphasizes beauty standards related to ageing and sun damage.[27,29,30] For instance, magazines targeting predominantly female audiences tend to feature advertisements for sun protection products, whereas magazines targeting predominantly male audiences have significantly fewer or no such advertisements.[31] This is reflected in our results whereby 44% of females compared to 36% of males report using sunscreen as a protection against accelerated skin ageing. The formulation of sunscreen can also have a negative impact on its use among

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males. The oily consistency of sunscreen and skin irritation have been reported to be main reasons deterring them from using sunscreen.[32]

Interestingly, protection against accelerated skin ageing was a better incentive to using sunscreen than prevention from skin cancer. This was particularly true in females (44% vs. 28%, respectively) and in participants from Asia (42% vs. 24%, respectively). It has been suggested that individuals may perceive themselves as more susceptible to developing wrinkles and age spots compared to cancer, potentially due to the higher prevalence and visibility of the former.[33] In one study performed on 35 female British students, aged 18–34, with the majority having a skin type of II, III or IV, viewing morphed UV-aged images of themselves motivated them to change either their sun protection and/or their UV exposure behaviours.[34] These findings imply that appearance-based interventions can play a substantial role in healthcare and educational settings concerning sun protection behaviour. It is also essential to highlight that the immediate effects of sun exposure can have a great impact on sun protective behaviour, with more than 50% reporting using sunscreen to avoid sunburn (up to 59% in Africa and 56% in Latin America).

One in five participants reported applying sunscreen to spend more time in the sun. This finding aligns with previous reports. Indeed, it has been demonstrated that sunscreen use leads to longer duration of sun exposure when sun exposure is intentional, but not when sun exposure is not intentional.[35] Sunburns tend to be paradoxically more frequent among sunscreen users.

The strength of our study was the large sample from an international representative sample across five continents. Nevertheless, Internet-based questionnaires are a well-accepted and reliable method for collecting quantitative data, and allows wider access to geographically distant participants, as achieved in our study. Another limitation is that the questionnaire did not ask about profession, particularly about outdoor professions such as construction workers, farmers or lifeguards who are more likely to be exposed to the sun throughout the day. Although the studied population was representative of the population in each country based on socio-economic-cultural level, academic, and climate levels, we did not perform a comparison based on these variables. Further studies in each country may help tailor specific recommendations to each country or region. For instance, in a large part of Africa, people do not have any access to the internet, often not even electricity, nor is there sunscreen on sale, in addition to the fact that sun exposure is not a problem, as the largest portion of the population contains sufficient melanin for natural protection, and in some other countries, the cult of tanning is part of local culture, for example in Brazil. Lastly, our study is limited by recall bias, particularly concerning the 12-month timeframe, and it was carried out in winter/early spring in the northern hemisphere and summer/early autumn in the southern hemisphere. The study does not.

CONCLUSIONS

People across the world fail to consistently avoid sun exposure and use sunscreen, despite recognizing the significance and potential repercussions of not abiding by sun protective measures. Forgetfulness and the challenges posed by time constraints seem to be the biggest barriers to proper adherence. This should prompt health professionals and health authorities to tailor their message in a way that resonates with those who require persuasion, ensuring it is effectively communicated to the target audience. Pricing sunscreens reasonably and developing formulations to reduce the inconvenience of their application are challenges that need to be addressed by the manufacturers.

AUTHOR CONTRIBUTIONS

Charles Taieb: Conceptualization, data curation, formal analysis, funding acquisition; investigation; methodology; project administration; supervision; writing – review and editing. Yaron Ben Hayoun: Formal analysis, project administration. Henry W. Lim, Marketa Saint Aroman, Charbel Skayen, Marie Aleth Richard, Bruno Halioua, Nuria Perez Cullell, Catherine Baissac Khaled Ezzedine: Conceptualization, methodology; writing – review and editing. Christina Bergqvist: Writing – original draft.

ACKNOWLEDGEMENTS

None.

FUNDING INFORMATION

This study was granted by the Patient Centricity Pierre Fabre.

CONFLICT OF INTEREST STATEMENT

Henry W. Lim has no conflicts of interest to declare in this study. Marketa Saint Aroman is employed by the Pierre Fabre Laboratory. Charbel Skayen has no conflicts of interest to declare in this study. Bruno Halioua has no conflicts of interest to declare in this study. Nuria Perez Cullell is employed by the Pierre Fabre Laboratory. Yaron Ben Hayoun has no conflicts of interest to declare in this study. Catherine Baissac is employed by the Pierre Fabre Laboratory. Christina Bergqvist has no conflicts of interest to declare in this study. Charles Taieb has no conflicts of interest to declare in this study. Marie-Aleth Richard has no conflicts of interest to declare in this study. Khaled Ezzedine has no conflicts of interest to declare in this study.

DATA AVAILABILITY STATEMENT

The data sets generated during and/or analysed during this study are available from the authors upon reasonable request.

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How to cite this article: Lim HW, Saint Aroman M, Skayem C, Halioua B, Perez Cullell N, Ben Hayoun Y, et al. Sun exposure and protection habits: Self-reported attitudes, knowledge and behaviours. J Eur Acad Dermatol Venereol. 2024;38:2024–2033. https://doi.org/10.1111/jdv.20080

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